

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	6a3a555ae5895b2c9759029d
Project Title	EduGenie - Google Gemini Powered Learning Assistant Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Give students a single AI-powered assistant that answers questions, explains concepts at the right depth, generates practice quizzes, summarizes notes, and builds a personalized learning roadmap - all backed by Google Gemini with an offline/local-model fallback. A FastAPI web application with five request/response modules (chat, explanation, quiz, summary, learning path), a Jinja2/HTML/CSS/JS frontend, JSON-file persistence for query/response history, and a Dockerized deployment path.
Scope	Learning Process, incorporating machine learning for a more robust and Students lack a single, adaptive tool that can explain a concept at their level, quiz them on it, summarize their notes, and tell them what to study next - they currently stitch this together from multiple generic tools.
Problem Statement	Solving this reduces study friction and time-to-understanding, gives students a low-cost self-testing mechanism, and helps them plan their learning path with less guesswork.
Description	Customer Satisfaction.
Impact	Use Google Gemini (with dynamic best-model resolution and retry/backoff) as the primary generative engine, prompted with strict JSON response schemas per feature, and a local HuggingFace LaMini-Flan-T5 model plus a built-in mock/demo mode as fallbacks so the app is always usable.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict CPU / GPU for running the local T5 model (CUDA auto-detected, uses GPU if available). RAM to hold the local LaMini-Flan-T5 model in memory when loaded (2-4 GB recommended).
Key Features	Disk space for model weight cache, Docker image, and JSON database files (2-3 GB). FastAPI, Uvicorn, Jinja2 template rendering engines. google-generativeai, transformers, torch, pydantic, python-dotenv, httpx, python-multipart, pytest.

	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	CPU / GPU for running local T5 (CUDA auto-detected)
Memory	RAM specifications	2-4 GB RAM recommended for local model loading
Storage	Disk space for data, models, and logs	2-3 GB for model weights cache and data logs
Software		
Frameworks	Python frameworks	FastAPI application server with Uvicorn
Libraries	Additional libraries	google-generativeai SDK, transformers, PyTorch, pydantic
Development Environment	IDE	Visual Studio Code, python-dotenv, Git/GitHub
Data		
Data	Source, size, format	Local structured JSON flat-file storage logs